ORF309/MAT380
Discrete Random Variables I: Expectations & Distributions

(pp.26-30)

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Agenda for today

- Discrete/Continuous random variables
- Expectation
- Distribution
  - Joint
  - Marginal
  - Examples

PSET1 due TODAY!

Two highly motivated and capable UCAs have joined the team.

Their office hours are:

- **Anany Kotawala: Thursdays 4:30 - 6:30pm**
- **Akash Bhowmick: Thursdays 6:30 - 8:30pm**

They will share the same room: 003 Sherrerd Hall

Discrete Random Variables

- **Definition**: A random variable $X: \Omega \to D$ is called **discrete** if the value set D is a finite or countable set
  - **Examples**:
    - The **number of students** in room 101 every Monday at 9:35AM
    - The **number of cars** that arrive at the parking lot at a given day
    - The **number of people** who will win the Mega Million jackpot this week

Continuous Random Variables

- **Definition**: A random variable $X: \Omega \to D$ is called **continuous** if the value set D is a NOT countable set
  - **Examples**:
    - The **speed of a car** passing outside the CS building → $D = (0,25]$
    - The **time** it takes to complete a lecture of ORF309 → $D = (0,53]$
    - The **gallons per mile** for a BMW Z4 → $\mathrm{D} = [25,33]$

Expectations

Expectations

- **Definition**: If $X: \Omega \to D$ is a **discrete r.v. (DRV)** and $f: D \to R$ is a value function, then the **expectation of** $\boldsymbol{f(X)}$ is defined as

$$E[f(X)] = \sum_{i \in D} f(i) P\{X = i\}$$

- **Remark**: if $D \subset R$ then we can choose $\boldsymbol{f(i) = i}$ and the expectation of X becomes:

$$E[f(X)] = \sum_{i \in D} i\, P\{X = i\}$$

- **Note**: this definition makes sense **only for DRV**, because for **continuous r.v. (CRV)** we have $\boldsymbol{P\{X = x\} = 0}$, for all $x \in D$.

Expectations

- **Example 1:**
  - Consider the DRV: $X =$**"suit of a card in a deck"**
  - **Modeling assumptions**: Each suit is equally likely → $P\{X = "any\ suit"\ \} = 1/4$
  - The value function $f: D \to \{1,2,3,4\}$, where $D = \{\spadesuit, \diamondsuit, \clubsuit, \heartsuit\}$
    - Think of the values 1, 2, 3, 4 as the number of $$$ you get if you get that suit
  - Then: $E[f(X)] = \sum_{x \in D} f(x)\, P\{X = x\} =$

$$f(\spadesuit)P\{X = \spadesuit\} + f(\diamondsuit)P\{X = \diamondsuit\} + f(\clubsuit)P\{X = \clubsuit\} + f(\heartsuit)P\{X = \heartsuit\} =$$

$$= 1\frac{1}{4} + 2\frac{1}{4} + 3\frac{1}{4} + 4\frac{1}{4} = \frac{10}{4} = \frac{5}{2}.$$

Expectations – Indicator Function

- **Example 2**: (very IMPORTANT for the rest of course!)
  - Assume A is an event. Then we can define the DRV $\mathbf{1_A}$ with values:

$$1_A(\omega) = \begin{cases} 1, if \ \omega \in A \\ 0, if \ \omega \notin A \end{cases}$$

  - The DRV is called the **indicator RV**.
  - Let's compute the **expectation** of 1_A

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  - Let's compute the **expectation** of 1_A

$$E[1_A] = 1P\{1_A = 1\} + 0P\{1_A = 0\} = P\{1_A = 1\} = P(A)$$

  - So, the expectation of $\mathbf{1_A}$ is the probability the event A

Indicator Function – Usefulness!

The indicator function allows us to represent the **number of successes in a series of trials** as a simple sum.

Example:

- Consider the DRV $\boldsymbol{X}$ representing the **number of heads** in n coin flips (**Successes**).
- Define the **events**: $\boldsymbol{X_i}$ = "the $\boldsymbol{i}$**-th** flip is **head**" for $i = 1,2,\dots,n$. ($\boldsymbol{i}$**-th success**)
- For each event $\boldsymbol{X_i}$ we can define the corresponding **indicator function** $\mathbf{1}_{\boldsymbol{X_i}}$ ($\boldsymbol{i}$**-th success or failure**)
- Then we can express $\boldsymbol{X}$ as the sum of the indicator function:

$$\boldsymbol{X} = \mathbf{1}_{\boldsymbol{X_1}} + \mathbf{1}_{\boldsymbol{X_2}} + \cdots + \mathbf{1}_{\boldsymbol{X_n}}$$

Distributions of DRVs

Distributions of DRVs

- **Definition**: If $X : \Omega \to D$ is a DRV, then its distribution is given by specifying the probabilities of all its outcomes $(P\{X = i\})_{i \in D}$

- **Remark**: The distribution of a DRV X is information needed to compute $E[f(X)]$ for any function f

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- **Definition**: Uniform Distribution
  - Let D be a finite set. The DRV $X : \Omega \to D$ for which

$$P\{X = i\} = \frac{1}{|D|}, \forall i \in D$$

is said to be the **Uniform Distribution** on D.

Notation: $\boldsymbol{X \sim Unif(D).}$

Distributions of DRVs

- **Remark**: We will use the information about the **distribution of RV** in the **Modeling Assumptions** of our **Probability Model**. This will allow to compute Expectations, Variances, etc.

- **Remark**: the distribution of a RV X provides information about X **only**, and not for any other RV Y.

  - **Example:**

    - $X =$"the number that comes out when **I** throw the die"

    - $Y =$"the number that comes out when **YOU** through the die"

      (no collusion! OR no conspiracy!)

    - If X, Y are **independent**, and *if I do not tell you what to do*, then the distribution of X **does not provide any information** about the distribution of Y

Joint Distribution

- Suppose you have two DRVs: $X:\Omega \to D_X$, $Y:\Omega \to D_Y$ and $f: D_X \times D_Y \to R$

- **Question 1**: How can we compute $\boldsymbol{E}[\boldsymbol{f}(\boldsymbol{X},\boldsymbol{Y})]$??

Joint Distribution

- Suppose you have two DRVs: $X : \Omega \to D_X$, $Y : \Omega \to D_Y$ and $f : D_X \times D_Y \to R$
- **Question 1**: How can we compute $\boldsymbol{E[f(X,Y)]}$??
  - We can define a **new** DRV: $Z : \Omega \to D_X \times D_Y$ such that $Z = (X, Y)$
  - Then the expectation of the new DRV Z is defined as

$$E[f(X,Y)] = \sum_{x \in D_X,\, y \in D_Y} f(x,y)\, \underbrace{P\{X = x, Y = y\}}_{\textit{\textbf{JOINT} distribution of } X,Y}$$

Joint Distribution

- **Definition**: the **Joint Distribution** of two DRVs $X : \Omega \to D_X$ and $Y : \Omega \to D_Y$ is specified by the probabilities $(P\{X = i, Y = j\})_{i \in D_X, j \in D_Y}$
- **Definition**: The distribution of the DRV X **only**, is called the **marginal distribution** of X (similarly for Y)
- **Remark**:
  - The Joint distribution of X, Y allows us to answer questions about **both** X and Y
  - The Marginal distribution of X (or Y) allows us to answer questions about X **only** (or Y only) when we know the joint distribution.

Joint vs Marginal distributions

- The Marginal distributions **do NOT determine** the Joint distribution. We need additional assumptions

  - Think of the "2 dice throwing experiment" where we know the Marginal distributions $P\{X = 1\} = \cdots = P\{X = 6\} = 1/6$ and $P\{Y = 1\} = \cdots = P\{Y = 6\} = 1/6$,

  - BUT the Joint distribution of $X \& Y$ needs to be computed.

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  - BUT the Joint distribution of $X\&Y$ needs to be computed.
- The Joint distribution **can be used to determine** the Marginal distributions.
  - If you know $(P\{X=x,Y=y\})_{x\in D_X, y\in D_Y}$ then you can determine the Marginal distribution of X by assuming that Y is already known (e.g., fixed to some value).

Joint vs Marginal distributions

- Assume that we know the Joint distribution $P\{X = i, Y = j\}$ and we want to compute the Marginal distribution $P\{X = i\}$.

- **CLAIM**: $P\{X = i\} = \sum_{j \in D_Y} P\{X = i, Y = j\}$ → We sum over all possible outcomes of Y

  - **Proof by illustration**:

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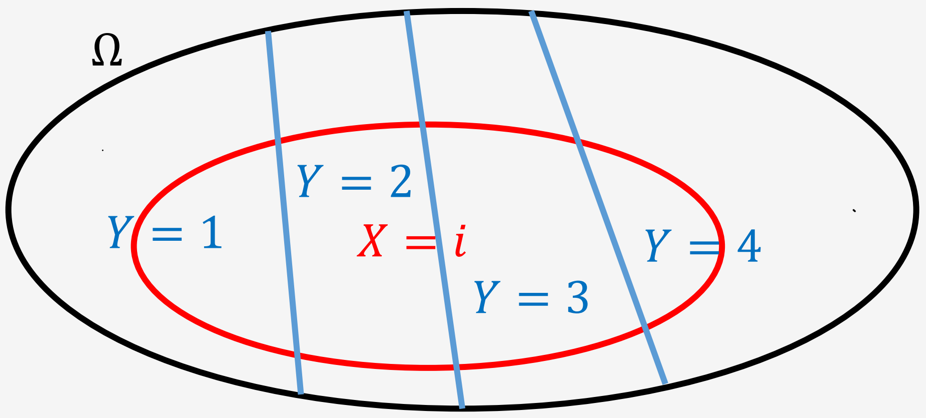

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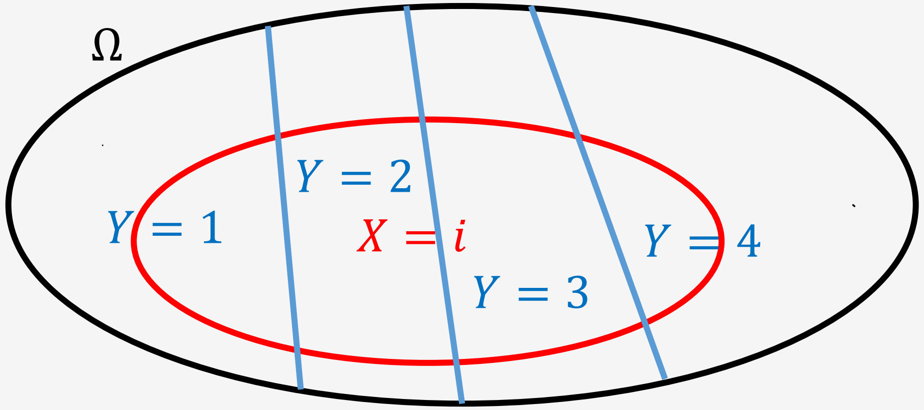

Assume the DRV: $Y : \Omega \to \{1,2,3,4\}$

We can write: $\Omega = \{Y = 1\} \cup \{Y = 2\} \cup \{Y = 3\} \cup \{Y = 4\}$

Also: $\{Y = i\} \cap \{Y = j\} = \emptyset, \forall i, j$ → mutually exclusive

Then: $\{X = i\} = \{X = i, Y = 1\} \cup \cdots \cup \{X = i, Y = 4\} \Rightarrow$

$$\Rightarrow P\{X = i\} = \sum_{j \in D_Y} P\{X = i, Y = j\}$$

Example: <u>Two fair dice</u>

- You are given the **joint distribution**: $P\{\, X = x, Y = y\} = \frac{1}{36}, \;\; \forall (x,y) \in \{(1,1),(1,2),\dots,(6,6)\}$
- You want to calculate the marginal distribution: $P\{X = x\}, \;\; \forall x \in \{1,2,3,4,5,6\}$
- **Note**: $\{X = x\} = \{X = x, Y = 1\} \cup \cdots \cup \{X = x, Y = 6\}$
- **Note also**: $\{X = x, Y = y_i\} \cap \left\{X = x, Y = y_j\right\} = \emptyset, \forall (x,y) \in \{(1,1),(1,2),\dots,(6,6)\}$
- **Taking probabilities**:

$$P\{X = x\} = \{X = x, Y = 1\} + \cdots + \{X = x, Y = 6\} = 6\frac{1}{36} = \frac{1}{6}, \forall x \in \{1,\dots,6\}$$

** Similarly for Y : $P\{Y = y\} = \frac{1}{6}, \forall y \in \{1,\dots,6\}$

- So, the probability distribution each of the dice is **uniform**.
- Note also that X, Y are **independent**, because $P\{X = x, Y = y\} = \frac{1}{36} = \frac{1}{6}\frac{1}{6} = P\{X = x\}P\{Y = y\}$

Remark: knowing that two dice are **individually fair** (their marginal distributions are uniform) **does not mean** they always can be **independent**

Consider this "**strange joint distribution**":

You have **two fair dice** but when you **throw them together** this is what you observe:

$$P\{X = x, Y = y\} = \begin{cases} \frac{1}{6}, if\ x = y \\ 0, if\ x \neq y \end{cases}$$

Marginal for $\boldsymbol{X}$: $P\{X = x\} = P\{X = x, Y = 1\} + \cdots + P\{X = x, Y = 6\} = \frac{1}{6} + 0 + \cdots + 0 = \frac{1}{6}, \forall x$

Marginal for $\boldsymbol{Y}$: $P\{Y = y\} = P\{X = 1, Y = y\} + \cdots + P\{X = 6, Y = y\} = \frac{1}{6} + 0 + \cdots + 0 = \frac{1}{6}, \forall y$

Are X and Y independent??

$$P\{X = 1, y = 1\} = \frac{1}{6} \neq \frac{1}{36} = P\{X = 1\}P\{Y = 1\}$$

Hence, X and Y are **NOT independent**!

Expectation – Joint and Marginal Distributions

- **Example**: You know the distributions of 2 DRVs X, Y and you want to compute $E[X+Y] = ??$

$$E[X+Y] = \sum_{x \in D_X} \sum_{y \in D_Y} (x+y)\, P\{X=x, Y=y\} =$$

$$= \sum_{x \in D_X} \sum_{y \in D_Y} x\, P\{X=x, Y=y\} + \sum_{x \in D_X} \sum_{y \in D_Y} y\, P\{X=x, Y=y\} =$$

$$\sum_{x \in D_X} x \underbrace{\sum_{y \in D_Y} P\{X=x, Y=y\}}_{\substack{\text{Marginal of } X \\ (\text{sum over } y\text{'s})}} + \sum_{y \in D_Y} y \underbrace{\sum_{x \in D_X} P\{X=x, Y=y\}}_{\substack{\text{Marginal of } Y \\ (\text{sum over } x\text{'s})}} = \sum_{x \in D_X} x\, P\{X=x\} + \sum_{y \in D_Y} y\, P\{Y=y\} =$$

$$= E[X] + E[Y]$$

Practice exercises/examples

(may not be covered during lectures but they are offered for your practice)

- **Example 1**: we toss two coins <u>once</u>. Let's assume that the 1st coin is blue and the 2nd is red. The sample space is defined as

$$\Omega = \{HH, HT, TH, TT\}$$

- For the first coin we have: $p_1 = P\{H\}$ and $1 - p_1 = P\{T\}, \quad 0 \leq p_1 \leq 1$

- For the second coin we have: $p_2 = P\{H\}$ and $1 - p_2 = P\{T\}, \quad 0 \leq p_2 \leq 1$

- The probabilities of each atomic event are

$$P\{HH\} = p_1 p_1, P\{HT\} = p_1(1 - p_2), P\{TH\} = (1 - p_1)p_2, P\{TT\} = (1 - p_1)(1 - p_2)$$

- We want to figure out whether the following events are **independent**:

  - $A = \{HH, HT\} = \{the\ 1st\ coin\ shows\ H\}$ and $B = \{HH, TH\} = \{the\ 2nd\ coin\ shows\ H\}$

- We need to compute the probabilities: $P(A), P(B), P(A \cap B)$ and show whether $P(A \cap B) = P(A)P(B)$.

- $P(A) = P(\{HH\} \cup \{HT\}) = P(\{HH\}) + P(\{HT\}) = p_1 p_2 + p_1(1 - p_2) = p_1$

- $P(B) = P(\{HH\} \cup \{TH\}) = P(\{HH\}) + P(\{TH\}) = p_1 p_2 + (1 - p_1)p_2 = p_2$

- $P(A \cap B) = P(\{HH\}) = p_1 p_2 = P(A)P(B)$

- Hence $P(A \cap B) = P(A)P(B)$ and we showed that A, B are **independent events**.

- **Remark 1**: note that $A \cap B$ is a new event which basically expresses the statement: "the 1st coin shows H **and** the 2nd coin shows H". That's why we can write $A \cap B = \{HH\}$.

- **Remark 2**: you can also use Set Theory to derive the same result:

$$A \cap B = (\{HH\} \cup \{HT\}) \cap (\{HH\} \cup \{TH\}) = (\{HH\} \cap \{HH\}) \cup (\{HH\} \cap \{TH\}) \cup (\{HT\} \cap \{HH\}) \cup (\{HT\} \cap \{TH\}) = \{HH\} \cup \emptyset \cup \emptyset \cup \emptyset = \{HH\}$$

Independence

- **Exercise**: show that if A and B are independent events, then A^C and B^C are independent events too.

Indicator Function Properties

- **Exercise:** prove the following properties
  - $1_{A \cap B} = 1_A 1_B$
  - $1_{A \cup B} = 1_A + 1_B - 1_{A \cap B}$
  - $1_{A^c} = 1 - 1_A$